

Protocol for Frozen-Record Boundary-Twist Charge Redistribution in an Adaptive Chern Circuit

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¹*Class-A fermionic adaptive-circuit project*
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We specify a compact CPU pilot for diagnosing boundary-twist charge redistribution in the class-A adaptive free-fermion circuit. A zero-twist Born record is generated once and then imposed in independent static replays at a grid of twisted boundary conditions. Separate records are required for soft explicit interfaces and hard support-terminated walls because their active measurement sets differ. The primary geometry is a 16×20 cylinder evolved for $2N_y = 40$ perfect-correction cycles with a pure half-filled initial state, random serial measurement order, $n_{\text{shell}} = 1$, and complex128 arithmetic. Both increasing and decreasing sweeps use 17 points with the signed origin convention $\phi_j^{(\sigma)} = -\sigma 10^{-7} + \sigma 2\pi j/16$. Only the final left/right regional charges are evaluated. The same imposed record fixes the total injected charge at every twist, so twist-dependent redistribution is isolated by subtracting the corresponding infinitesimally displaced origin. The experiment is a conditional static-response probe, not a quantized monitored pump: neighboring twists are not dynamically connected and the 2π endpoint must close by a large gauge transformation. This note fixes the scientific contract, offset rationale, charge estimator, numerical checks, parallel execution, resumability, output schema, and claim boundary; production plots and numerical conclusions are intentionally deferred.

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I. PURPOSE AND STATUS

The adaptive preparation protocol of Bhuiyan, Pan, and Jian measures localized overcomplete-Wannier (OW)

occupations and applies outcome-dependent fermionic feedback without postselection [1]. A twist around the periodic y direction probes transverse redistribution between two x -separated interfaces, in the spirit of the Laughlin

cylinder argument [2, 3]. Twisted boundary conditions also provide a real-space formulation that does not require translation symmetry [4, 5].

This document describes a deliberately small, reproducible numerical pilot. It records what is being calculated and how its correctness is established; it does not report or interpret production curves. The pilot is intended to answer a narrow implementation question before a more expensive ensemble or online-ramp calculation: if the stochastic word and therefore the net feedback source are held fixed, does changing a static boundary twist alter the final partition of charge between the two walls?

Four related constructions must not be conflated:

1. The exact flattened-Hamiltonian benchmark continuously follows occupied wall branches and can exhibit nearly integer endpoint transfer.
2. The present frozen-record experiment applies one imposed stochastic word in independent static simulations.
3. An online monitored pump changes the twist during one uninterrupted Born-sampled trajectory and requires cycle-resolved source subtraction.
4. Modular charge spreading evolves a localized packet in modular time; it is a propagation diagnostic rather than physical monitored time.

Only item 2 is implemented here.

II. SCIENTIFIC CONTRACT

A. Geometry and OW measurements

The physical system has $N_x \times N_y = 16 \times 20$ unit cells with two physical orbitals per cell. A domain-wall mass profile places a topological Chern-insulator region with $\alpha_1 = 1$ on $x \in [4, 12]$ and a matched trivial exterior with $\alpha_2 = 30$. The OW basis uses trial family X and $n_{\text{shell}} = 1$. Distinct OW stabilizers need not commute, and the OW states form an overcomplete localized frame rather than an orthonormal Wannier basis.

The two wall implementations are

- *soft*: `dw_truncation=False` and `meas_slab_only=False`, retaining full OW tails and measurements throughout the system;
- *hard*: `dw_truncation=True` and `meas_slab_only=True`, terminating OW support and restricting measurements to the topological slab.

These are distinct stochastic protocols, so each receives its own reference record. They are compared at identical geometry, cycle count, initialization, ordering, filling, and arithmetic precision.

B. State representation and dynamics

The canonical CPU engine stores the centered covariance Γ . The physical single-particle correlation matrix is

$$G = \frac{\mathbf{1} + \Gamma}{2}, \quad G_{ij} = \text{Tr}(\hat{\rho} \hat{c}_i^\dagger \hat{c}_j), \quad (1)$$

where $\hat{c}_i$ is a physical lattice fermion and $\mathbf{1}$ is the single-particle identity. The runner calls `classA.U1FGTN.run_markov_circuit` for every sampled reference and every forced replay. It neither copies the Markov-cycle loop nor stores a cycle-resolved covariance history.

Each trajectory begins from the same deterministic root seed and evolves for 40 complete cycles with pure-state rank-one covariance updates. The reference calculation records the ordered site identifiers, measurement outcomes, conditional correction targets, and selected-branch probabilities required for an exact replay.

III. REFERENCE RECORD AND STATIC REPLAY

A. Zero-twist record

Let $\mathbf{m}_a$ denote the complete ordered word for arm $a \in \{\text{soft}, \text{hard}\}$. It is generated by an ordinary Born-sampled run at $\phi = 0$. The runner saves

$$\{G_{a,\text{init}}(0), G_{a,\text{final}}(0), \mathbf{m}_a, \Delta N_{\text{inj},a}\}. \quad (2)$$

Before any twist tasks are admitted, a forced replay at $\phi = 0$ must reproduce the sampled final covariance. The preregistered discrepancy is

$$\epsilon_{0,a} = \frac{\|\Gamma_{a,\text{replay}}(0) - \Gamma_{a,\text{sampled}}(0)\|_F}{2N_x N_y} \leq 10^{-12}. \quad (3)$$

This check validates the record serialization, event ordering, initial state, and deterministic replay path on a common input.

TABLE I. Locked pilot configuration.

geometry	$N_x \times N_y = 16 \times 20$
domain wall	$x \in [4, 12], \alpha_1 = 1, \alpha_2 = 30$
OW basis	trial X , $n_{\text{shell}} = 1$
initial state	pure random, half filling
measurement order	random serial
feedback	perfect correction, no postselection
duration	$40 = 2N_y$ physical cycles
arithmetic	complex128
twist points	17 per direction
independent records	one soft, one hard
canonical entry point	<code>classA.U1FGTN.run_markov_circuit</code>

B. Signed twist grid

Write $\sigma = +1$ for increasing/counterclockwise flux and $\sigma = -1$ for decreasing/clockwise flux. The 17-point grid is

$$\boxed{\phi_j^{(\sigma)} = -\sigma\varepsilon + \sigma\frac{2\pi j}{16}}, \quad j = 0, \dots, 16, \quad \varepsilon = 10^{-7}. \quad (4)$$

Thus the increasing sweep starts at -10^{-7} and the decreasing sweep starts at $+10^{-7}$. The same exactly zero-twist reference word is used for both directions; the offset changes the static evaluation point, not the sampled record.

At each $\phi_j^{(\sigma)}$, the model and OW projectors are rebuilt at that static twist. The reference initial state is placed in the uniform gauge,

$$G_{a,\text{init}}(\phi) = U_\phi G_{a,\text{init}}(0) U_\phi^\dagger, \quad [U_\phi]_{xy\mu, xy\mu} = e^{i\phi y/N_y}, \quad (5)$$

and $\mathbf{m}_a$ is imposed through all 40 cycles. Every twist is a fresh static simulation. There is no state transfer from point j to $j + 1$.

C. Why the 10^{-7} offset is used

The offset is not a physical mass, broadening, or Hamiltonian regulator. It resolves a convention at the exponentially small avoided crossing of the two physical-wall branches at zero flux. In the exact flattened-Hamiltonian 20×40 benchmark, the closest levels at $\phi = 0$ are approximately

$$E = \pm 5.49 \times 10^{-10} \quad (\text{soft}), \quad (6)$$

$$E = \pm 7.51 \times 10^{-10} \quad (\text{hard}). \quad (7)$$

Starting infinitesimally to one side selects an unambiguous wall-localized branch. The quantization-favoring choice has sign opposite to the winding direction, which is precisely Eq. (4). The exact continued benchmark then gives

Using the opposite side of the crossing gives magnitudes 0.98397184 for the soft wall and 0.98080969 for the hard wall. Those deficits reflect the small initial weight on the opposite wall, not an inadequate flux grid. Nearly integer transfer belongs to continuously tracked exact-Hamiltonian branches. Because the present pilot re-evaluates each twist independently, quantization is explicitly excluded from its acceptance criteria.

TABLE II. Continued exact-Hamiltonian reference at 20×40 . These values motivate the signed origin convention; they are not acceptance targets for the frozen-record circuit.

wall	increasing from -10^{-7}	decreasing from $+10^{-7}$
Soft	+0.9999999976	-0.9999999924
Hard	+0.9999999987	-0.9999999994

IV. ENDPOINT CHARGE OBSERVABLE

A. Fixed injected charge

For a measurement event e , let $m_e \in \{0, 1\}$ be the measured occupation and $s_e \in \{0, 1\}$ the perfect-correction target. The imposed word fixes the net particle source

$$\Delta N_{\text{inj},a} = \sum_{e \in \mathbf{m}_a} (s_e - m_e). \quad (8)$$

This integer is identical at every twist and for both orientations within the same arm. Each replay must satisfy global charge continuity,

$$N_{\text{final}}(\phi) - N_{\text{initial}}(\phi) = \Delta N_{\text{inj},a} \quad (9)$$

to absolute tolerance 10^{-9} . The pilot deliberately measures charge only after the entire record has been applied; no intermediate $\Delta N(\phi)$ is assigned.

B. Left-right partition

The cylinder is divided into two disjoint half-system regions,

$$R_L : x = 0, \dots, 7, \quad R_R : x = 8, \dots, 15, \quad (10)$$

$$R_L + R_R = \mathbf{1}. \quad (11)$$

The endpoint regional charges are

$$N_b(\phi) = \text{tr}[R_b G_{a,\text{final}}(\phi)], \quad b = L, R. \quad (12)$$

Each orientation is referenced to its own infinitesimally displaced origin,

$$\Delta N_b^{(\sigma)}(\phi_j) = N_b(\phi_j^{(\sigma)}) - N_b(\phi_0^{(\sigma)}), \quad (13)$$

$$q_{\text{wall}}^{(\sigma)}(\phi_j) = \frac{\Delta N_R^{(\sigma)}(\phi_j) - \Delta N_L^{(\sigma)}(\phi_j)}{2}. \quad (14)$$

No regional source term is subtracted. The global source is already identical at all twists, so it cancels when the same-word origin is subtracted. The residual check

$$\Delta N_L^{(\sigma)} + \Delta N_R^{(\sigma)} = 0 \quad (15)$$

must hold to 10^{-9} .

C. Large-gauge closure

The points $j = 0$ and $j = 16$ differ by a signed 2π flux and are related by a large gauge transformation. Endpoint covariances are retained only for these two points, and the closure diagnostic is

$$\epsilon_{\text{cl}}^{(a,\sigma)} = \frac{\|U_{-\sigma 2\pi} \Gamma_a(\phi_{16}^{(\sigma)}) U_{-\sigma 2\pi}^\dagger - \Gamma_a(\phi_0^{(\sigma)})\|_F}{2N_x N_y}. \quad (16)$$

The preregistered ceiling is 10^{-8} . A valid static replay should therefore return $q_{\text{wall}}(2\pi)$ to zero. This closure is a correctness condition and is the principal reason that the final point must not be read as a quantized pump endpoint.

V. EXECUTION AND RESUMABILITY

A. Parallel task decomposition

The full scan contains

$$2 \text{ walls} \times 2 \text{ directions} \times 17 \text{ twists} = 68 \quad (17)$$

deterministic replay tasks, plus two reference-record tasks. The soft and hard references are sampled in parallel. Once both are verified, the 68 static replays are distributed across a process pool. The default host allocation uses CPU cores 0–55, 56 workers, and one BLAS thread per worker. Independent twists of the same record are safe to parallelize because they share immutable serialized inputs and no evolving state.

A single parent `tqdm` bar reports completed and failed twist points. Worker stdout is suppressed to prevent 56 copies of the canonical engine progress stream from making the log unreadable. The campaign is launched inside `tmux` so terminal detachment does not terminate the calculation.

B. Completion-based resume

Each twist task writes one compressed NPZ followed by one completion JSON. The completion file binds

- the deterministic task and configuration hashes,
- the exact wall-specific record SHA-256,
- the result filename and byte count, and
- the result SHA-256.

On restart, a point is skipped only if the result and completion files both exist and every declared identity, size, and checksum matches. Missing, partial, or mismatched pairs are rerun. The expensive scientific unit is one static replay, which is comfortably shorter than the repository threshold for intra-task state checkpointing; therefore no cycle-level checkpoint, pointer protocol, lease, Drive API, or distributed coordination layer is present.

The reference records use the same principle but bind three files: the compressed trajectory word, the parent-state NPZ, and a completion JSON. A campaign-identity file prevents a changed configuration or source from being silently resumed under the same identifier.

VI. SAVED PRODUCTS AND PLANNED ANALYSIS

The runner saves only what is required to reconstruct and audit the endpoint response:

Full covariance histories are deliberately excluded. After all verified points exist, the planned two-column diagnostic figure will show: (a) soft-wall ΔN_L and ΔN_R ; (b) hard-wall ΔN_L and ΔN_R ; (c) q_{wall} for both arms and directions; and (d) $|\Delta N_L + \Delta N_R|$ against the preregistered tolerance. No error bars belong on the pilot curves because the 17 points are deterministic functions of one shared word, not independent samples.

The first analysis pass must report, without threshold tuning,

1. the two reference injections and zero-twist replay errors;
2. the minimum selected-branch probability;
3. maximum global-continuity and regional-balance residuals;
4. four large-gauge closure errors;
5. the peak and half-loop wall response for each arm and direction; and
6. any disagreement between clockwise and counter-clockwise static evaluations at gauge-equivalent twists.

VII. ALLOWED INTERPRETATION

If the structural checks pass and a finite $q_{\text{wall}}(\phi)$ is observed, the strongest justified pilot statement is:

For each of two explicit perfect-correction trajectories, the final Gaussian state conditioned on the realized measurement word re-distributes charge between the two wall basins

TABLE III. Pilot data products.

reference word	ordered measurement and correction events, gzip JSON
reference state	initial/final covariance, injection, replay diagnostics
twist point	N_L, N_R, N_{tot} , x profile, likelihood diagnostics
endpoint only completion	full covariance at $j = 0, 16$ for closure task/record/config identity, bytes, SHA-256
aggregate manifest	CSV and NPZ over all 68 points configuration, sources, residual maxima, closure values

as a function of static boundary twist, while the global feedback injection remains fixed.

This is a trajectory-conditioned response statement. It is not an ensemble mean because there is only one independent record per wall. It is not a quantized pump because neighboring twists are independent static recalculations and the 2π endpoint closes. It is not evidence for a physical current because the twist does not vary during monitored time.

A paper-level conditional-response result would repeat the full construction for independent records and use the trajectory as the resampling unit. A physical monitored-pump claim would instead require one online Born trajectory with a cycle-dependent twist, live charge accounting, and explicit source-subtracted regional balance. The present pilot should be used to validate twist plumbing, record replay, wall partitioning, charge continuity, and sign conventions before either extension.

VIII. REPRODUCIBILITY

The self-contained project is `00_WORKSPACE/CURRENT/frozen_record_flux_charge_pilot/`. Its locked configuration is `campaign_config.v1.json`; the campaign runner is `run_campaign.py`; and the local launcher is `launch_tmux.sh`. The nominal campaign identifier is `N16x20_frozen_flux_charge_v1`,

with root seed 2026090202. The result root is `results/N16x20_frozen_flux_charge_v1/`.

The implementation has unit and small end-to-end CPU tests for the exact 68-task expansion, signed twist convention, injection parser, regional charge partition, checksum failure, deterministic serial/parallel agreement, and completion-based resume. Production data are not part of this protocol note; when incorporated later, their manifest and generated analysis inputs must be treated as the numerical authority rather than values copied manually into the manuscript.

IX. CONCLUSION

This protocol isolates a useful but deliberately limited observable: the endpoint charge response of a fixed monitored-circuit word to a static boundary twist. Its two wall constructions, signed 10^{-7} origins, fixed injection identity, half-system partition, and large-gauge closure test are fully specified before inspection of production curves. The design is small, parallel, and completion-resumable, while avoiding engineering layers that do not protect scientific correctness. Its eventual output will decide whether the conditional response is promising enough to justify an ensemble extension; it cannot by itself establish quantized or dynamical pumping.

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